

How is a signal entering a listener's ears converted into a linguistic message?

The conversion of sound into a linguistic message is a multi-stage process involving physiological and neurological steps, as described in the speech chain.

1. **Acoustic Level:** Sound waves first enter the listener's ear. The outer ear directs these waves toward the eardrum.
2. **Physiological Level (Ear):**
 - The **middle ear** transforms the sound pressure into mechanical motion.
 - The **inner ear** then converts these mechanical vibrations into electrical nerve impulses. Specifically, a traveling wave moves down the **Basilar Membrane (BM)**, which acts like a series of filters, causing maximal vibration at different locations depending on the sound's frequency.
 - **Hair cells** on the BM detect this motion and convert it into neural firings, which are collected by the auditory nerve.
3. **Physiological to Linguistic Level (Brain):**
 - The auditory nerve transmits these impulses to the brain.
 - The brain performs **neural processing** on these signals to create the perceived sound.
 - Finally, at the **linguistic level**, the brain interprets this sound as a sequence of linguistic units and understands the idea being communicated.

How do we discriminate gender, emotion, etc.? What are pitch, loudness, and intonation?

While the document does not explicitly detail the discrimination process, it defines the key acoustic properties we use to make these judgments. We infer characteristics like gender and emotional state from variations in these properties.

- **Pitch:** Pitch is the perceptual property that depends on the **fundamental frequency** of a sound wave's vibration. A higher frequency results in a higher, more shrill pitch. For instance, a woman's voice generally has a higher pitch than a man's.
- **Loudness:** Loudness corresponds to the **amplitude or intensity** of the sound wave. A wave with a larger amplitude is perceived as louder. Loudness is measured in decibels (dB).
- **Intonation:** Intonation is the **variation in pitch** across an utterance, creating its melodic pattern. Along with stress and rhythm, intonation conveys expressive meanings such as surprise, anger, or wariness.
- **Rhythm:** The document notes that rhythm is a component involved in intonation but does not provide a separate definition.

How is speech perceived in noise?

The provided text does not directly answer how speech is perceived in noise. However, it describes auditory mechanisms that are fundamental to this process. One key concept mentioned is **auditory masking**, where tones or noise can obscure sounds within the same critical frequency band. The auditory system's ability to analyze sound via critical bands in the cochlea allows the brain to separate different frequency components, which is a crucial first step in distinguishing speech from background noise.

What is the selective attention (cocktail party example) mechanism?

The provided document **does not contain information** explaining the selective attention mechanism or the cocktail party effect. This question is posed in the "Self-check" section but is not answered in the text.

What are the different auditory models?

The document describes and lists several models that attempt to simulate human auditory perception.

- **Perceptual Linear Prediction (PLP):** This model incorporates several perceptual effects, including analysis on a non-linear Bark frequency scale, adjustments for the ear's unequal sensitivity (equal-loudness curve), and compression of sound intensity to match perceived loudness.
- **Seneff Auditory Model:** This is a comprehensive three-stage model that simulates the response of the cochlea and hair cells. It includes a critical band filter bank, a hair cell synapse model, and detectors that extract features like formants and phonetic transitions.
- **Other Models:** The document also briefly mentions **Lyon's Cochlear model** and the **Ensemble Interval Histogram model** by name.